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SEMICONDUCTOR CHIP AND LIGHT-EMITTING DEVICE

Abstract

A semiconductor chip includes a semiconductor stack, a passivation layer, a first electrode and a second electrode. The semiconductor stack includes a top surface, a bottom surface opposite to the top surface, and a sidewall between the top surface and the bottom surface. The passivation layer covers the sidewall of the semiconductor stack. The first electrode and the second electrode are respectively located on the top surface and the bottom surface of the semiconductor stack. When viewing from the bottom surface, a plan-view contour of the passivation layer surrounds a plan-view contour of the semiconductor stack and includes a plurality of protruding portions protrude outwards from the plan-view contour of the passivation layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a divisional application of U.S. application Ser. No. 17/384,812, filed Jul. 26, 2021, which claims priority to China Application serial number 202110289497.3, filed Mar. 18, 2021, all of which are herein incorporated by reference in their entireties.

BACKGROUND

Field of Invention

[0002] The present disclosure relates to a semiconductor chip and a light-emitting device.

Description of Related Art

[0003] In recent years, various novel displays have gradually become popular. These displays are mainly developed towards a direction of increasing resolution and saving energy. A micro light-emitting diode (μ LED) is one of the important types of displays in development.

[0004] Micro light-emitting diodes have reduced the size of conventional light-emitting diodes to below about 100 micron, or even to a magnitude of tens of micron. The number of the LEDs in the same area significantly increases at this magnitude, so the yield of the LEDs transferred from a substrate to a display substrate should reach above 99%. With techniques of recent processes, this mass transfer still faces several problems to be solved.

SUMMARY

[0005] Some embodiments of the present disclosure provide a semiconductor chip including a semiconductor stack, a passivation layer, a first electrode and a second electrode. The semiconductor stack includes a top surface, a bottom surface opposite to the top surface, and a sidewall between the top surface and the bottom surface. The passivation layer covers the sidewall of the semiconductor stack. The first electrode and the second electrode are respectively located on the top surface and the bottom surface of the semiconductor stack. When viewing from the bottom surface, a plan-view contour of the passivation layer surrounds a plan-view contour of the semiconductor stack and includes a plurality of protruding portions protruding outwards therefrom.

[0006] Some embodiments of the present disclosure provide a semiconductor chip including a semiconductor stack and a passivation layer. The semiconductor stack includes a top surface, a bottom surface opposite to the top surface, and a sidewall between the top surface and the bottom surface. The passivation layer covers the sidewall of the semiconductor stack and includes a first protruding portion and a second protruding portion. When viewing from the bottom surface, in a plan-view of the passivation layer, the first protruding portion outwardly protrudes with a first length and the second protruding portion outwardly protrudes with a second length different from the first length.

[0007] Some embodiments of the present disclosure provide a light-emitting device including a substrate, a plurality of light-emitting diode chips, and a passivation layer. The plurality of light-emitting diode chips is arranged in a first direction on the substrate, in which one of the light-emitting diode chips includes a semiconductor stack including a top surface, a bottom surface opposite to the top surface, and a sidewall between the top surface and the bottom surface. The passivation layer covers the top surface and the sidewall of the semiconductor stack. The passivation layer includes a protruding portion extending in an extending direction with respect to the first direction by a first angle, and the first angle is an acute angle.

[0008] It is to be understood that both the foregoing general description and the following detailed

description are by examples, and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

[0010] FIGS. **1-9** illustrate cross-section views of a semiconductor chip in the intermediate manufacturing process in accordance with some embodiments of the present disclosure.

[0011] FIGS. **10A-10B** illustrate cross-section views of the semiconductor chip in accordance with some embodiments of the present disclosure.

[0012] FIGS. **11A-11D** illustrate plan views of the semiconductor chip from the bottom surface in accordance with some embodiments of the present disclosure.

[0013] FIG. **12** illustrates a cross-section view of a semiconductor chip in accordance with some embodiments of the present disclosure.

[0014] FIGS. **13A-13B** illustrate cross-section views of a light-emitting device including light-emitting units and a control unit in accordance with some embodiments of the present disclosure.

[0015] FIG. **13C** illustrates a top view of FIGS. **13A-13B**.

[0016] FIGS. **14-21** illustrate cross-section views of a semiconductor chip in the intermediate manufacturing process in accordance with other embodiments of the present disclosure.

[0017] FIG. **22** illustrates a cross-section view of the semiconductor chip in accordance with some embodiments of the present disclosure.

[0018] FIG. **23A** illustrates a plan view of the semiconductor chip in the intermediate manufacturing process before transfer in accordance with some embodiments of the present disclosure.

[0019] FIGS. **23B-23C** illustrate top views of a light-emitting device using semiconductor chips in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0020] Reference will now be made in detail to the present embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.

[0021] In various embodiments, description is made with reference to figures. However, certain embodiments may be practiced without one or more of these specific details, or in combination with other known methods and configurations. In the following description, numerous specific details are set forth, such as specific configurations, dimensions and processes, etc., in order to provide a thorough understanding of the present disclosure. In other instances, well-known semiconductor processes and manufacturing techniques have not been described in particular detail in order to not unnecessarily obscure the present disclosure. Reference throughout this specification to “one embodiment,” “an embodiment”, “some embodiments” or the like means that a particular feature, structure, configuration, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Thus, the appearances of the phrase “in one embodiment,” “in an embodiment”, “in some embodiments” or the like in various places throughout this specification are not necessarily referring to the same embodiment of the disclosure. Furthermore, the particular features, structures, configurations, or characteristics may be combined in any suitable manner in one or more embodiments.

[0022] The terms “over,” “to,” “between” and “on” as used herein may refer to a relative position of one layer with respect to other layers. One layer “over” or “on” another layer or bonded “to” another layer may be directly in contact with the other layer or may have one or more intervening

layers. One layer “between” layers may be directly in contact with the layers or may have one or more intervening layers.

[0023] Embodiments of the present disclosure can increase the number of chips produced per wafer. Specifically, the location of tether structures of the chips may be changed that tether structures of the chips are placed at the corners of the chips. This may decrease the distance between the chips, so that more chips are produced per wafer. Further, this also may reduce the area of tether structures remaining on the transferred chips and improve the lighting efficiency of the chips.

[0024] FIGS. **1-9** illustrate cross-section views of a semiconductor chip in the intermediate manufacturing process in accordance with some embodiments of the present disclosure. In FIG. **1**, a first substrate **110** is provided. The first substrate **110** may be any suitable materials. In some embodiments, the first substrate **110** is a sapphire substrate. Then a semiconductor stack **120** may be deposited on the first substrate **110**. The semiconductor stack **120** may include n-type/p-type semiconductor layers made of different materials, or the same material doped with n-type/p-type impurities, for example, a first semiconductor layer **121**, a second semiconductor layer **122** and an optionally formed third semiconductor layer **123** as shown in FIG. **1**. In some embodiments, the semiconductor stack **120** may also include an active layer **124** between the first semiconductor layer **121** and the second semiconductor layer **122**. In some embodiments, a transparent conductive layer **125** may be deposited on the semiconductor stack **120**.

[0025] In some embodiments, the first semiconductor layer **121** is a n-type doped GaN and the second semiconductor layer **122** is a p-type doped GaN, or vice versa. In addition, the third semiconductor layer **123** may be an undoped GaN as a buffer layer between the first semiconductor layer **121** and the first substrate **110**. The active layer **124** may be GaN added with quantum well structures and/or quantum dot structures, therefore when current flowing through the active layer **124**, the active layer **124** may emit light with specific wavelength. The transparent conductive layer **125** may be made of indium tin oxide (ITO) or other suitable materials.

[0026] As shown in FIG. **2**, trenches **134** and **135** are formed in the transparent conductive layer **125** and the semiconductor stack **120**. In some embodiments, an etching process, such as a dry etching process, may be used to form the trenches **134** and **135**. For example, a first etching process is performed in the transparent conductive layer **125** and the semiconductor stack **120** to form a shallow and wide trench **134**. A second etching process is then performed to form a deep and narrow trench **135** in the first semiconductor layer **121** and the semiconductor layer **123**. In this etching process, the trenches **134** and **135** partially divide the semiconductor stack **120** into a tether structure **126** and a patterned semiconductor stack **127** (i.e. the bottom of the trenches **134** and **135** is spaced apart from the first substrate **110** by a distance), and the trench **135** defines sidewalls **128** (such as the plan-view contour **129** in FIGS. **11A-11D**) of the patterned semiconductor stack **127**. The sidewalls **128** may define the shape of the subsequently formed semiconductor chip **100**. The location of the tether structure **126** is also determined in this operation and will be discussed in FIGS. **11A-11D** in greater detail. In some embodiments, a distance, which is used to form the electrode holes and electrodes in subsequent processes, is between the boundaries of the trenches **134** and **135**.

[0027] As shown in FIG. **3**, a passivation layer **140** is conformally deposited on the transparent conductive layer **125** and the semiconductor stack **120**. Thus, the outer contour of the passivation layer **140** (such as the plan-view contour **145** in FIGS. **11A-11D**) is only slightly larger than the outer contour of the semiconductor chip **100** (i.e. a range defined by the sidewalls **128**). The passivation layer **140** may be used to protect the exposed surfaces of the semiconductor stack **120** to prevent from the penetration of external particles and water vapor in the environment, and ensure that there is no electrical connection between the first semiconductor layer **121**, the second semiconductor layer **122** and the subsequently formed conductive layers except through the electrode holes. In some embodiments, the passivation layer **140** may be an insulator layer.

[0028] Next, a first electrode hole **142** and a second electrode hole **144** are formed in the passivation layer **140**. The first electrode hole **142** and the second electrode hole **144** are used to respectively expose the second semiconductor layer **122** (or the transparent conductive layer **125** which electrically connects to the second semiconductor layer **122**) and the first semiconductor layer **121**.

[0029] Next, as shown in FIG. 4, a first electrode **152** and a second electrode **154** are respectively formed in the first electrode hole **142** and the second electrode hole **144**. The first electrode **152** and the second electrode **154** are both on the same side of the semiconductor stack **120**. The first electrode **152** and the second electrode **154** are respectively in contact with the second semiconductor layer **122** (or the transparent conductive layer **125** which electrically connects to the second semiconductor layer **122**) and the first semiconductor layer **121** to provide electrical connection between the semiconductor chip **100** and the circuit board of the display device in the subsequent application.

[0030] As shown in FIG. 5, a sacrificial layer **160** is formed on the passivation layer **140**, the first electrode **152** and the second electrode **154**. After forming the sacrificial layer **160**, a recess **162** is formed over the tether structure **126** by performing an etching process or other suitable methods. Moreover, the bottom portion of the recess **162** exposes a portion of the passivation layer **140** over the tether structure **126**.

[0031] As shown in FIG. 6, a supporting structure layer **172**, a bonding layer **174** and a second substrate **176** are sequentially deposited on the sacrificial layer **160**, and the supporting structure layer **172** is in contact with the passivation layer **140** over the tether structure **126** through the recess **162** (as shown in FIG. 5). In some embodiments, the supporting structure layer **172** is made of SiO₂ and may be used to support the tether structure **126** and the patterned semiconductor stack **127** during the subsequent device-transferring process. The bonding layer **174** is a metal layer interconnecting semiconductor layer and the second substrate **176**, and the second substrate **176** may be the substrate similar to the first substrate **110**.

[0032] As shown in FIG. 7, the structure as shown in FIG. 6 is flipped, and the first substrate **110** is removed. In some embodiments, an etching process, laser-lift-off process may be used to remove the first substrate **110**. In some embodiments, after removing the first substrate **110**, the exposed surface of the third semiconductor layer **123** is a rough surface.

[0033] As shown in FIG. 8, a portion of the semiconductor stack **120** is removed until a portion of the surface of the passivation layer **140** between the tether structure **126** and the semiconductor stack **127** is exposed. The bottom surface **127a** of the patterned semiconductor stack **127** is also exposed. The bottom surface **127a** may remain rough, and may be flat, too. The bottom surface **127a** herein refers to the surface farther from the active layer **124** in the semiconductor stack **127**. In other words, in this operation, the passivation layer **140** will not be removed. Any suitable methods, such as chemical mechanical polishing, etching process, etc., may be used to remove the portion of the semiconductor stack **120**. After removing the portion of the semiconductor stack **120**, the manufacturing process of the semiconductor chip **100** (as shown in FIG. 9) is finished. In some embodiments, the semiconductor chip **100** may be used as a light-emitting diode (LED) chip in the subsequent application.

[0034] As shown in FIG. 9, the sacrificial layer **160** is removed. Any suitable methods, such as chemical etching, may be used to remove the sacrificial layer **160**. Therefore, the sacrificial layer **160** is able to be removed under the situation that the semiconductor chip **100** is not damaged. After removing the sacrificial layer **160**, the semiconductor chip **100** is attached to the tether structure **126** by the passivation layer **140**. Subsequently, the semiconductor chip **100** is transferred by using transfer techniques. The semiconductor chip **100** (as shown in FIG. 10A) includes the semiconductor stack **127**, the passivation layer **140**, the first electrode **152** and the second electrode **154**. During transferring the semiconductor chip **100**, a transfer apparatus (such as stamp) is in contact with the semiconductor chip **100** from the bottom surface **127a**, and then the transfer

apparatus removes the semiconductor chip **100**. Meanwhile, the tether structure **126** is still fixed on the supporting structure layer **172**, so the operation of removing separates the semiconductor chip **100** from the tether structure **126**. After being separated, the semiconductor chip **100** is transferred to the device in subsequent application, such as a circuit board (such as the substrate **310** in FIG. **13A**).

[0035] FIG. **10A** illustrates a cross-section view of the semiconductor chip **100** in accordance with some embodiments of the present disclosure. The semiconductor stack **127** of the semiconductor chip **100** has a bottom surface **127a** and a top surface **127b**, and sidewalls **128** are between the bottom surface **127a** and the top surface **127b**. The top surface **127b** is closer to the active layer **124** compared with the bottom surface **127a**. Because the semiconductor chip **100** is directly removed from the tether structure **126** by a transfer apparatus, the passivation layer **140** fractures at a connection part between the tether structure **126** and the semiconductor stack **127**, forming a protruding portion **182** with a fracture surface **184**. The protruding portion **182** and the bottom surface **127a** are substantially coplanar. In some other embodiments, a concave-and-convex structure **186** may be formed on the bottom surface **127a**, as shown in FIG. **10B**. If the bottom surface **127a** has the concave-and-convex structure **186**, total internal reflection of light may be reduced to improve light extraction.

[0036] FIGS. **11A-11D** illustrate plan-views of the semiconductor chip **100** from the bottom surface **127a** in accordance with some embodiments of the present disclosure. In FIG. **11A**, the semiconductor stack **127** of the semiconductor chip **100** has a plan-view contour **129** with multiple edges **130**. The plan-view contour **129** has four edges **130** if the plan-view contour **129** is a quadrilateral. However, in some other embodiments, the plan-view contour **129** may be a triangle or polygon with more than four sides. Two adjacent edges **130** together define a corner **131**. In FIGS. **11A-11D**, the plan-view contour **129** has four corners **131**.

[0037] The passivation layer **140** of the semiconductor chip **100** also has a plan-view contour **145**. The shape of the plan-view contour **145** may be substantially the same as the shape of the plan-view contour **129**. For example, in FIGS. **11A-11D**, the plan-view contour **129** is a quadrilateral, and the plan-view contour **145** is also a quadrilateral. The plan-view contour **145** has multiple edges **146**. Two adjacent edges **146** together define a corner **147**. The corners **131** of the plan-view contour **129** are substantially aligned with the corners **147** of the plan-view contour **145**. The protruding portions **182**, as discussed above, are adjacent the corners **131** (i.e. at the corners **147**), and the protruding portions **182** protrude outwards from the plan-view contour **145** of the passivation layer **140**.

[0038] As discussed above, during the transfer after the manufacturing process in FIG. **9**, the passivation layer **140** fractures at the connection part between the tether structure **126** and the semiconductor chip **100**. The fractured passivation layer **140** is the protruding portion **182**, so the fracture part of the protruding portion **182** has the fracture surface **184**. The fracture surface **184** herein is a section randomly formed due to the fracture and the formation of the fracture surface **184** is related to removal strength and angles. Therefore, each of the protruding portions **182** has the fracture surface **184** with different shapes and each of the protruding portions **182** may also have different extension lengths. For example, in FIG. **11A**, two protruding portions **182** across corners may have the fracture surfaces **184** with different shapes and/or with different extension lengths. Similarly, in FIGS. **11C-11D**, each of the protruding portions **182** may have the fracture surfaces **184** with different shapes and/or with different extension lengths.

[0039] In addition, in FIG. **11A**, it is shown that the tether structures **126** (as shown in FIG. **9**) are originally disposed at the corners **147** of the semiconductor chip **100**. The tether structures **126** barely take up space of wafer, because the tether structures **126** are at the corners **147**. Therefore, adjacent semiconductor chips **100** are arranged closely when forming the semiconductor chips **100**, thereby increasing the number of the semiconductor chips **100** produced per wafer. Further, the area of the tether structures **126** remaining on the transferred semiconductor chips **100** (i.e. the

protruding portions **182**) may be reduced. This may effectively reduce the effect on light pattern of devices resulting from the remaining tether structures, and enhance the lighting performance of the semiconductor chips **100**.

[0040] In some embodiments, the semiconductor chips **100** have lengths and widths, and the widths of the protruding portions **182** are not greater than $\frac{1}{3}$ of the widths of the semiconductor chips **100**. The widths of the protruding portions **182** basically depend on materials of the passivation layer **140**. However, if the widths of the protruding portions **182** are too large, it is unfavorable for the separation of the semiconductor chips **100** and the tether structures **126**. If the widths of the protruding portions **182** are too small, the passivation layer **140** may not be able to support the semiconductor chip **100** after removing the sacrificial layer **160**.

[0041] Although FIG. **11A** illustrates that the semiconductor chip **100** has two protruding portions **182**, the number of the protruding portions **182** is not limited to two. As shown in FIGS. **11B-11D**, FIGS. **11B**, **11C** and **11D** respectively illustrate that the semiconductor chip **100** has one, three and four protruding portions **182**. In addition, each two of the non-adjacent corners **147** may form a diagonal, such as diagonal connecting lines **L1**, **L2** shown in FIGS. **11A-11D**. The ends of the diagonal connecting lines may be the middle point of the boundary of the corner **147** without the protruding portion **182**, or the middle point of the boundary of the protruding portion **182**. For example, the diagonal connecting line may be a line connecting the middle points of two fracture surfaces **184**, such as the connecting line **L1** shown in FIG. **11A**. The diagonal connecting line may be a line connecting the middle points of two corners **147**, such as the connecting line **L2** shown in FIG. **11A**. The diagonal connecting lines may be a line connecting the middle points of a corner **147** and a fracture surface **184**, such as the connecting line **L1** shown in FIG. **11B**. Due to the existence of the protruding portion **182**, the diagonal connecting lines **L1**, **L2** are not in equal lengths in each of the semiconductor chips **100**. This situation is present that when removing the semiconductor chip **100** from the tether structure **126** (as shown in FIG. **9**), the portions which connect to the tether structures **126** are not under the same forces, thereby causing the protruding portions **182** to be formed with different lengths.

[0042] The semiconductor chip in the present disclosure may not be limited to the LED chip discussed above. FIG. **12** illustrates a cross-section view of a semiconductor chip **200** in accordance with some embodiments of the present disclosure. In FIG. **12**, the semiconductor chip **200** may be an integrated circuit (IC) chip, and includes a semiconductor stack **227**, a passivation layer **240** and a first electrode **250**. The passivation layer **240** conformally covers the semiconductor stack **227**, and the first electrode **250** penetrates the passivation layer to be connected to the semiconductor stack **227**. Further, the passivation layer **240** includes a protruding portion **282** with a fracture surface **284**. The difference between the semiconductor chip **200** and the semiconductor chip **100** is the composition of the semiconductor stacks. In some embodiments, the semiconductor stack **227** of the semiconductor chip **200** may include silicon layer, etc. A metal layer, interconnections, devices (such as transistors, resistors, and capacitors, etc.), or other suitable structures may be formed in the semiconductor stack **227**, and the first electrode **250** may electrically connect to the devices in the semiconductor stack **227**.

[0043] FIG. **13A** illustrates a cross-section view of a light-emitting device **300** in accordance with some embodiments of the present disclosure. The light-emitting device **300** includes a substrate **310**, light-emitting units **320** and a control unit **330**. The light-emitting units **320** and the control unit **330** are arranged on the substrate **310**. The substrate **310** may be a circuit board to provide electricity for the light-emitting units **320** and the control unit **330**.

[0044] The light-emitting units **320** include LED chips **322**. The LED chips **322** may be the semiconductor chips **100** as discussed before. As the embodiments shown in FIG. **13A**, the LED chips **322** may be blue light LED chips. In some embodiments, the light-emitting units **320** are light-emitting units **320B** able to emit blue light.

[0045] In other embodiments, the light-emitting units **320** are light-emitting units **320R** able to emit

red light and light-emitting units **320G** able to emit green light. The light-emitting units **320R** and the light-emitting units **320G** may further include conversion layers and filters. In each of the light-emitting units **320R** and the light-emitting units **320G**, the conversion layers **324** with quantum dots or phosphor materials are disposed on the corresponding LED chips **322** to convert the light emitted from the LED chips **322** into light with different wavelengths, such as red light or green light. The filters **326** for different colors are disposed on the corresponding conversion layer to filter light with specific wavelength. For example, a red light conversion layer **324R** may convert blue light emitted from the LED chips **322** into red light to emit red light through a red light filter **326R**. A green light conversion layer **324G** may convert blue light emitted from the LED chips **322** into green light to emit green light through a green light filter **326G**. Apart from the blue LED chips, the LED chips in the light-emitting units **320R** and **320G** may be other suitable chips, such as ultra violet LED chips. Opaque structures **328** are between each two of the light-emitting units **320** to ensure lights emitted from the light-emitting units **320** emit towards a desired direction and prevent from crosstalk between the adjacent light-emitting units **320**. The red light, green light and blue light respectively emitted from the light-emitting units **320R**, **320G**, and **320B** may be mixed to form white light and may be used in subsequent applications.

[0046] The control unit **330** includes an IC chip **332** electrically connecting to the light-emitting units **320**. The IC chip **332** may be the semiconductor chips **200** as discussed before. The control unit **330** may be used to control the light-emitting units **320**, such as the brightness and the switch of the light-emitting units **320**.

[0047] In some embodiments, an encapsulation material **340** is filled on the LED chips **322** and the IC chip **332**. The encapsulation material **340** is a transparent encapsulation material used for preventing the LED chips **322**, the IC chip **332** and the substrate **310** from being damaged (such as water vapor or oxygen).

[0048] The light-emitting units **320** and the control units **330** may be arranged in any suitable patterns. In some embodiments, the arrangement of the light-emitting units **320** (such as light-emitting units **320R**, **320G** and **320B**) and the control units **330** is shown as FIG. **13C**. However, it is noted that the arrangement of the light-emitting units **320** and the control unit **330** is not limited to the pattern shown in FIG. **13C**. For example, the light-emitting units **320R**, **320G** and **320B** and the control unit **330** may be arranged in a same row in order, as shown in FIG. **13A**.

[0049] In some other embodiments, the light-emitting device **300** is shown as FIG. **13B**. The LED chips **322** of the light-emitting units **320R** and **320G** are respectively a red LED chip and a green LED chip. Therefore, the light-emitting units **320R** and **320G** may not respectively include conversion layers **324R** and **324G** but emit the red light and the green light.

[0050] FIGS. **14-21** illustrate cross-section views of a semiconductor chip in the intermediate manufacturing process in accordance of other embodiments of the present disclosure, wherein FIGS. **14-21** are cross-section views along line A-A' in FIG. **23A**. In FIG. **14**, a first substrate **410** is provided, and a semiconductor stack **420** and a transparent conductive layer **425** may be deposited on the first substrate **410**. Detailed discussion related to the first substrate **410**, the semiconductor stack **420**, and the transparent conductive layer **425** are similar to or same as the discussion related to FIG. **1**, so the discussion of the embodiments is not repeated herein.

[0051] As shown in FIG. **15**, trenches **434** and **435** are formed in the semiconductor stack **420** to define the location of the tether structure **426** and the patterned semiconductor stack **427**, further defining the shape of the patterned semiconductor stack **427**, such as a circle (as shown in FIG. **23A**) or a quadrilateral. The distance between the boundaries of the trenches **434** and **435** may be shortened compared to the trenches **134** and **135** in FIG. **2**. Detailed discussion related to the manufacturing process in FIG. **15** is similar to or same as the discussion related to FIG. **2**, so the discussion of the embodiments is not repeated herein.

[0052] As shown in FIG. **16**, a passivation layer **440** is conformally deposited on the semiconductor stack **420** and the transparent conductive layer **425**. Detailed discussion related to the passivation

layer **440** is similar to or same as the discussion related to FIG. 3, so the discussion of the embodiments is not repeated herein. A first electrode hole **442** is then formed in the passivation layer **440** to expose the semiconductor stack **420** (or the transparent conductive layer **425**). After forming the first electrode hole **442**, a first electrode **452** is formed in the first electrode hole **442**. [0053] As shown in FIGS. 17-19, a sacrificial layer **460**, a supporting structure layer **472**, a bonding layer **474** and the second substrate **476** are formed over the passivation layer **440** and the first electrode **452**. The structure shown in FIG. 18 is then flipped and the first substrate **410** and a portion of the semiconductor stack **420** are removed to form a bottom surface **427a**. Detailed description related to the sacrificial layer **460**, the semiconductor layer **472**, the bonding layer **474**, the second substrate **476** and the manufacturing process are similar to or same as the discussion related to FIGS. 5-8, so the discussion of the embodiments is not repeated herein.

[0054] In some embodiments, as shown in FIG. 20, a second electrode **480** is formed on the bottom surface **427a** of the semiconductor stack **427**. The second electrode **480** and the first electrode **452** are on the opposite sides of the semiconductor stack **420** In some other embodiments, the second electrode **480** may not be formed on the semiconductor stack **420**. In this operation, a complete structure of a semiconductor chip **400** is formed.

[0055] As shown in FIG. 21, the sacrificial layer **460** is removed and the semiconductor chip **400** is transferred to the devices in subsequent applications. The manufacturing process in FIG. 21 is similar to or the same as the discussion related to FIGS. 5-9, so the discussion of the embodiments is not repeated herein.

[0056] The transferred semiconductor chip **400** is shown as FIG. 22. The semiconductor stack **427** of the semiconductor chip **400** has a bottom surface **427a** and a top surface **427b**, and sidewalls **428** are between the bottom surface **427a** and the top surface **427b**. Also, the passivation layer **440** of the semiconductor chip **400** includes a protruding portion **482** with a fracture surface **484**. The difference between the semiconductor chip **400** and the semiconductor chip **100** in FIG. 10A (or in FIG. 10B) is that the second electrode **480** and the first electrode **452** are on the opposite sides of the semiconductor stack **427**, while the first electrode **152** and the second electrode **154** are at the same side of the semiconductor stack **127**. Other detailed description related to the semiconductor chip **400** is similar to or same as the discussion related to FIG. 10A, so the discussion of the embodiments is not repeated herein.

[0057] FIG. 23A illustrates a plan view of the semiconductor chip **400** in the intermediate manufacturing process before transfer in accordance with some embodiments of the present disclosure. In FIG. 23A, the tether structures **426** on the supporting structure layer **472** provide support for the semiconductor chips **400**. The passivation layers **440** of the semiconductor chips **400** have circular plan-view contours and are attached to the tether structures **426** by the connecting portions **448**. The semiconductor chips **400** are arranged on the supporting structure layer **472** in a first direction X, and a first extension direction D1 of the connecting portions **448** and the first direction X together define a first angle a1. In some embodiments, the first angle a1 may be in a range from about 10° to about 80°, such as about 45°.

[0058] FIG. 23B illustrates a top view of a light-emitting device **500** in accordance with some embodiments of the present disclosure. The semiconductor chips **400** are transferred to the substrate **510**, as shown in FIG. 23B. During the transfer, the transfer apparatus does not change the direction (such as rotate) of the semiconductor chips **400**, so the semiconductor chips **400** are still arranged in the first direction X after transferring to the substrate **510**. The protruding portions **482** formed after the transfer still extend in the first extension direction D1, and the angle between the first extension direction D1 and the first direction X is also the first angle a1. After or before transferring the semiconductor chips **400**, the semiconductor chips **200** shown in FIG. 12 may be transferred, and the semiconductor chips **200** are arranged in the first direction X, electrically connecting to the semiconductor chips **400**. The extension direction of the protruding portions **282** of the semiconductor chips **200** is the second extension direction D2, and the second extension

direction D2 and the first direction X together define a second angle α_2 . In some embodiments, the second angle α_2 may be in a range from about 10° to about 80°, such as about 45°. After transferring the semiconductor chips **400** and the semiconductor chips **200**, other structures similar to FIGS. **13A-13B**, such as transfer layers, filters, transparent encapsulation materials and opaque structures, may be subsequently formed to produce display devices.

[0059] Although FIG. **23B** illustrates that the semiconductor chip **400** after the transfer has two protruding portions **482**, the number of the protruding portions **482** is not limited. In some embodiments, the number of the connecting portions of the semiconductor chip **400** may be adjusted, thereby adjusting the number of the protruding portions **482** of the semiconductor chip **400**, as shown in FIG. **23C**. Further, in some embodiments, a portion of the semiconductor chips **400** and **200** are not in the same pattern as shown in FIG. **23B**; for example, the protruding portions of a portion of the semiconductor chips **400** and **200** may be parallel to the first direction X.

[0060] Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

[0061] It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A semiconductor chip, comprising: a semiconductor stack comprising a top surface, a bottom surface opposite to the top surface, and a sidewall between the top surface and the bottom surface; a passivation layer covering the sidewall of the semiconductor stack; and a first electrode and a second electrode respectively located on the top surface and the bottom surface of the semiconductor stack; wherein when viewing from the bottom surface, a plan-view contour of the passivation layer surrounds a plan-view contour of the semiconductor stack and comprises a plurality of protruding portions protruding outwards therefrom.
2. The semiconductor chip of claim 1, wherein one of the plurality of protruding portions of the passivation layer has a fracture surface.
3. The semiconductor chip of claim 1, wherein the bottom surface has a concave-and-convex structure.
4. The semiconductor chip of claim 1, wherein two of the plurality of protruding portions are located symmetrically to the plan-view contour of the passivation layer.
5. The semiconductor chip of claim 1, wherein one of the plurality of protruding portions comprises a length protruding outwards different from a length of another one of the plurality of protruding portions protruding outwards.
6. The semiconductor chip of claim 5, wherein the one of the plurality of protruding portions and the another one of the plurality of protruding portions outwardly protrude in different directions.
7. The semiconductor chip of claim 1, wherein the plan-view contour of the semiconductor stack is a polygon shape.
8. The semiconductor chip of claim 1, wherein the plan-view contour of the semiconductor stack is a circle shape.
9. The semiconductor chip of claim 1, further comprising a transparent conductive layer between the semiconductor stack and the first electrode.
10. The semiconductor chip of claim 1, wherein the passivation layer is devoid of contacting the second electrode.
11. A light-emitting device, comprising: a substrate; a plurality of light-emitting units on the substrate, the light-emitting units emitting red light, green light or blue light respectively and an

- opaque structure being between two adjacent ones of the light-emitting units, wherein one of the light-emitting units comprises a semiconductor chip of claim 1; and a control unit on the substrate, wherein the control unit is electrically connected to the one of the light-emitting units.
- 12.** A semiconductor chip, comprising: a semiconductor stack comprising a top surface, a bottom surface opposite to the top surface, and a sidewall between the top surface and the bottom surface; a passivation layer covering the sidewall of the semiconductor stack and comprising a first protruding portion and a second protruding portion; and wherein when viewing from the bottom surface, in a plan-view contour of the passivation layer, the first protruding portion outwardly protrudes with a first length and the second protruding portion outwardly protrudes with a second length different from the first length.
- 13.** The semiconductor chip of claim 12, further comprising a first electrode and a second electrode which are on opposite sides of the semiconductor stack.
- 14.** The semiconductor chip of claim 13, further comprising a transparent conductive layer between the semiconductor stack and the first electrode.
- 15.** The semiconductor chip of claim 13, wherein the passivation layer is devoid of contacting the second electrode.
- 16.** The semiconductor chip of claim 13, wherein the passivation layer covers the top surface of the semiconductor stack, and the first electrode penetrates the passivation layer to electrically connect to the semiconductor stack.
- 17.** The semiconductor chip of claim 12, wherein a plan-view contour of the semiconductor stack is a circle shape.
- 18.** A light-emitting device, comprising: a substrate; a plurality of light-emitting units on the substrate, the light-emitting units emitting red light, green light or blue light respectively and an opaque structure being between two adjacent ones of the light-emitting units, wherein one of the light-emitting units comprises a semiconductor chip of claim 12; and a control unit on the substrate; wherein the control unit electrically connects to the one of the light-emitting units.
- 19.** A light-emitting device, comprising: a substrate; a plurality of light-emitting diode chips arranged in a first direction on the substrate, wherein one of the light-emitting diode chips comprises: a semiconductor stack comprising a top surface, a bottom surface opposite to the top surface, and a sidewall between the top surface and the bottom surface; and a passivation layer covering the top surface and the sidewall of the semiconductor stack; wherein the passivation layer comprises a protruding portion extending in an extending direction with respect to the first direction by a first angle, and the first angle is an acute angle.
- 20.** The light-emitting device of claim 19, wherein the protruding portion of the passivation layer has a fracture surface.
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